FoodGuard — Restricted Substances & Health Hazards Dataset

Clinical Toxicology, Target Organ Pathologies & Multi-Jurisdictional Regulatory Reference Matrix

1. Executive Overview & Problem Statement

Modern packaged food and agricultural supply chains frequently encounter chemical substances that have faced stringent legal bans or statutory restrictions across major international food safety authorities (FSSAI India, US FDA, EFSA EU). This dataset codifies the precise physiological and toxicological problems faced by each restricted substance, including cellular DNA fragmentation, endocrine disruption, organ-specific carcinogenesis, and severe pediatric behavioral effects.

2. Health Problems & Toxicological Profiles Table

Code	Substance	Primary Organ	Problems Faced (Clinical Hazards)	Banning / Restriction Status
INS 171	**Titanium Dioxide** Titanium dioxide (TiO2 nanoparticle...	**Cellular DNA & Gastrointestinal Tract**	Genotoxicity (DNA strand breaks, chromosomal aberration), systemic nanoparticle bioaccumulation in intestinal mucosa, liver, and spleen; unable to establish a safe Acceptable Daily Intake (ADI).	BANNED in European Union (2022); Permitted in USA & India with limitations
INS 127	**Erythrosine / FD&C; Red No. 3** Disodium 2,4,5,7-tetraiodofluoresce...	**Endocrine System & Thyroid Gland**	Thyroid follicular cell hypertrophy and adenomas/carcinogenesis; hyperthyroidism; pediatric neurobehavioral hyperactivity; iodine overload.	REVOKED by US FDA for ingested cosmetics/lakes; Banned in US for cosmetics; Under federal phase-out in food; RESTRICTED in EU solely to cocktail/candied cherries; Permitted with limits in India
INS 927a	**Azodicarbonamide (ADA)** 1,1'-Azobisformamide...	**Respiratory System & Cellular Metabolism**	Severe respiratory sensitization, occupational asthma, allergic rhinitis; thermal decomposition creates carcinogenic semicarbazide (SEM) and urethane (ethyl carbamate).	BANNED in European Union, United Kingdom, and Australia; Permitted up to 45 ppm in USA and India
INS 924a	**Potassium Bromate** Potassium bromate (KBrO3)...	**Renal (Kidneys), Thyroid & Auditory System**	Classified as an IARC Group 2B probable human carcinogen; induces renal cell tumors, thyroid follicular adenocarcinomas, peritoneal mesotheliomas, and ototoxicity (hearing impairment).	STRICTLY BANNED in India (FSSAI 2016), European Union, Canada, UK, China, and Brazil; Permitted in USA with advisory limit
BVO	**Brominated Vegetable Oil (BVO)** Brominated triglyceride fatty acids...	**Thyroid Gland, Heart Muscle & Hepatic Tissues**	Bioaccumulation of elemental bromine in adipose tissue and vital organs; myocardial degeneration; thyroid follicular cell hyperplasia; reproductive neurotoxicity; bromism symptoms (headache, fatigue, memory loss).	REVOKED / BANNED by US FDA (effective 2024); Banned in European Union, India, and Japan
INS 102	**Tartrazine / FD&C; Yellow No. 5** Trisodium 5-hydroxy-1-(4-sulfonatop...	**Central Nervous System & Immune/Respiratory System**	Pediatric neurobehavioral hyperactivity, Attention Deficit Hyperactivity Disorder (ADHD) aggravation; cross-reactive hypersensitivity in aspirin-sensitive asthmatics (severe bronchoconstriction and urticaria).	PERMITTED with strict limits in India & USA; MANDATORY WARNING LABEL in European Union ('May have an adverse effect on activity and attention in children')
INS 110	**Sunset Yellow FCF / FD&C; Yellow No. 6** Disodium 6-hydroxy-5-[(4-sulfopheny...	**Neurological & Immune Systems**	Increased motor restlessness and inattention in children; allergic rhinitis, angioedema, contact dermatitis; potential gastric irritation in sensitive individuals.	PERMITTED with quantitative limits in India & USA; EU MANDATORY WARNING LABEL regarding pediatric hyperactivity; ADI lowered by EFSA

INS 122	**Carmoisine / Azorubine** Disodium 4-hydroxy-3-[(4-sulfonato-...	**Respiratory & Dermatological Systems**	Acute asthmatic episodes in aspirin-intolerant patients; generalized urticaria; pediatric behavioral disruption; potential genotoxic risk in in-vitro assays.	BANNED / NOT AUTHORIZED in USA and Canada; Permitted with limits in India; Mandatory warning label in EU
INS 124	**Ponceau 4R / Cochineal Red A** Trisodium 1-(4-sulfo-1-naphthylazo)...	**Hepatic & Immune Systems**	Exacerbation of asthma, chronic urticaria, and angioedema; hyperactivity in preschool and school-age children; suspected renal DNA damage at high exposures.	BANNED / NOT AUTHORIZED in USA; Permitted with limits in India; Mandatory hyperactivity warning label in EU; ADI significantly reduced by EFSA
INS 104	**Quinoline Yellow** Disodium 2-(1,3-dioxo-2,3-dihydro-1...	**Neurological & Dermatological Systems**	Contact sensitization, dermatitis, asthma exacerbation; behavioral restlessness and loss of concentration in pediatric cohorts.	BANNED / NOT AUTHORIZED in USA and Canada; Permitted with limits in India and European Union (with warning label)
INS 143	**Fast Green FCF / FD&C; Green No. 3** Disodium 2-{[4-[ethyl(3-sulfonatobe...	**Cellular Integrity & Gastrointestinal Tract**	Poorly absorbed in intestine; associated with sarcomas at injection sites in animal studies; potential mutagenic responses in mammalian cell systems.	BANNED / NOT AUTHORIZED in European Union; Permitted with limits in USA and India
INS 952	**Sodium Cyclamate** Sodium cyclohexylsulfamate...	**Urinary Bladder & Male Reproductive System**	Banned in USA following animal bioassays indicating bladder carcinogenesis when combined with saccharin; metabolite cyclohexylamine causes testicular atrophy in rodents.	BANNED in USA since 1969; Permitted with quantitative limits in EU, Canada, and India
INS 320	**Butylated Hydroxyanisole (BHA)** 2-tert-butyl-4-methoxyphenol and 3-...	**Endocrine System & Gastrointestinal Mucosa**	Induces forestomach papillomas and squamous cell carcinomas in rodent bioassays; disrupts thyroid and sex steroid hormone homeostasis; potential immune system alterations.	RESTRICTED with strict maximum thresholds globally; Listed as suspected endocrine disruptor and reasonably anticipated human carcinogen in NTP Report
INS 319	**Tertiary Butylhydroquinone (TBHQ)** 2-(1,1-Dimethylethyl)-1,4-benzenedi...	**Immune System, Cellular DNA & Liver**	Suppresses T-cell and natural killer cell immune surveillance; precursor to DNA adduct formation at supra-nutritional doses; reported neurotoxic symptoms (vision disturbance, ringing in ears) during accidental acute exposure.	RESTRICTED: Maximum permissible limit of 200 mg/kg globally; Prohibited or not approved in infant foods; High dosages cytotoxic
ADULT-SR	**Sudan Dyes (Sudan I, II, III, IV)** 1-(Phenylazo)-2-naphthol and relate...	**Liver (Hepatic), Bladder & Genomic DNA**	Classified by IARC as potent genotoxic carcinogens; induce malignant hepatic carcinomas, urinary bladder tumors, and bone marrow chromosomal breakage.	STRICTLY ILLEGAL INDUSTRIAL ADULTERANT: Prohibited in all foodstuffs worldwide (Zero Tolerance)
ADULT-MY	**Metanil Yellow** Sodium 3-[4-(phenylamino)phenylazo]...	**Brain (Central Nervous System), Testes & Liver**	Severe depletion of brain monoamine neurotransmitters (dopamine, serotonin, noradrenaline); testicular atrophy, reduced sperm motility; acute hepatocellular necrosis.	STRICTLY ILLEGAL INDUSTRIAL ADULTERANT: Banned from food consumption under the Food Safety and Standards Act
ADULT-RB	**Rhodamine B** N-[9-(2-carboxyphenyl)-6-(diethylam...	**Cellular DNA, Liver, Kidneys & Thyroid**	Proven genotoxicity, chromosomal aberrations in peripheral lymphocytes; liver enlargement; acute renal toxicity; potential carcinogenic transformation in long-term exposures.	STRICTLY ILLEGAL INDUSTRIAL ADULTERANT: Industrial fluorescent dye prohibited in all food by FSSAI and WHO
ADULT-MEL	**Melamine** 1,3,5-Triazine-2,4,6-triamine...	**Renal System (Kidneys)**	Acute nephrotoxicity, renal tubular blockage, formation of insoluble kidney stones (melamine-cyanurate complexes), acute renal failure, and pediatric mortality.	STRICTLY ILLEGAL ADULTERANT: Worldwide zero-tolerance threshold in fluid milk; max 1 mg/kg in infant formula
ADULT-FORM	**Formalin / Formaldehyde** Aqueous solution of formaldehyde (C...	**Gastrointestinal Tract & Respiratory Epithelium**	Corrosive ulcerations of the stomach and duodenum, severe vomiting and hematemesis; nasopharyngeal carcinoma, leukemia; irreversible protein cross-linking.	STRICTLY ILLEGAL INDUSTRIAL CHEMICAL: IARC Group 1 known human carcinogen, completely prohibited in food

ADULT-ARG	**Argemone Oil (Sanguinarine)** Argemone mexicana seed oil containi...	**Cardiovascular System (Capillaries) & Eyes**	Causes Epidemic Dropsy (severe bilateral lower-limb edema, cardiac failure, dyspnea), glaucoma, optic nerve degeneration leading to permanent blindness, and hepatomegaly.	STRICTLY PROHIBITED TOXIC ADULTERANT: Zero tolerance under FSSAI Regulations

3. Academic Project Evaluation & Sign-off

Student Name / Roll: Final Year Engineering Team	**Department:** Computer Science & Engineering
Dataset Records: 20 Core Toxicological Profiles	**Academic Year:** 2026
Project Mentor: _____	**Signature / Approval:** _____